

Salem Toral Dynamics: Primitive and Cumulative Orbit Fluctuations

HCS-C397 theorem and reproducibility package

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Abstract

An explicit integral quartic family gives mixing symplectic toral maps with a nonexpansive central plane. We determine every finite fixed group, primitive cycle count, return multiplier and the complete rational Artin–Mazur zeta function, retaining the sign separating cardinalities from Lefschetz indices. Normalized primitive counts have an arcsine limiting law on $[0, 4]$. Cumulative counts have a different explicit nonconstant oscillatory prefactor, also with an arcsine distribution. Neither result requires a Diophantine lower bound for the unit-circle conjugates. These are complete source reconstructions of classical quasihyperbolic mechanisms, with independent exact arithmetic and numerical regression. Integral Salem arithmetic is not a prime-to-orbit correspondence, and no target spectral identity or literature priority is claimed.

Keywords: Salem number; toral automorphism; primitive orbit; dynamical zeta; homoclinic group; arcsine law.

中文摘要

本文研究一族整系数四次环面自同构，完整给出固定点群、本原轨道、返回稳定性和有理动力学函数。系统具有混合性和辛结构，但中心方向并不扩张。本原计数与累计计数分别呈现可显式描述的反正弦极限分布；累计轨道增长不能用一个常数前因子概括，证明不依赖单位圆根的丢番图下界。经典理论归属明确，整系数算术及完整源系统证明均不等于目标素数轨道或零点对应。

关键词：萨勒姆数；环面自同构；本原轨道；动力学函数；同宿群；反正弦律。

1 A nonexpansive integral source

Fix an integer $a \geq 1$, initially $a \neq 2$, and let

$$P_a(X) = X^4 - aX^3 - X^2 - aX + 1, \quad A_a = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & a \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & a \end{pmatrix}.$$

The source is $x \mapsto A_a x$ on $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$, with one iterate as its clock and normalized Haar measure. Classical accounts of quasihyperbolic dynamics include Lind [1]; Waddington [2] studies the nonconstant orbit asymptotics. The homoclinic obstruction is classical as well [3]. We prove this family's statements explicitly; the contribution here is complete source coverage and auditable conventions, not certified novelty. An abstract Salem stress polynomial in the project's Abel-law work does not supply these toral orbit and homoclinic theorems. Nor is this simply a hyperbolic cat-map parameter table.

Lemma 1. P_a is irreducible and has distinct roots $\lambda, \lambda^{-1}, e^{i\theta}, e^{-i\theta}$, where $\lambda > 1$ and $\theta/(2\pi)$ is irrational. The map is mixing, nonexpansive, symplectic and reversibly defined on the same torus.

Proof. Put $Z = X + X^{-1}$. Then $P_a(X)/X^2 = Z^2 - aZ - 3$, with

$$z_{\pm} = \frac{a \pm \sqrt{a^2 + 12}}{2}, \quad z_+ > 2, \quad -2 < z_- < 0, \quad 2 \cos \theta = z_-.$$

The only possible rational roots, ± 1 , give $1 \mp 2a \neq 0$. A factorization must therefore use two monic integral quadratics. If their constants are $+1$, their linear coefficients b, c satisfy $b + c = -a$, $bc = -3$, forcing $a = 2$ for positive a . If both constants are -1 , the cubic and linear coefficients have opposite signs, forcing $a = 0$. Gauss's lemma proves irreducibility in the stated range. A root of unity would then force every conjugate onto the unit circle, contrary to $\lambda > 1$.

The determinant of A_a is one. The integral form and reversor are

$$\Omega_a = \begin{pmatrix} 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ -a & -1 & 0 & 0 \end{pmatrix}, \quad R_a = [e_1, A_a^{-1}e_1, A_a^{-2}e_1, A_a^{-3}e_1].$$

Direct multiplication gives $\det \Omega_a = 1$, $A_a^\top \Omega_a A_a = \Omega_a$ and $R_a^\top \Omega_a R_a = -\Omega_a$. In $\mathbb{Q}[X]/(P_a)$ the integral involution $X \mapsto X^{-1}$ has matrix R_a , so $R_a^2 = I$ and $R_a A_a R_a = A_a^{-1}$.

Haar measure is preserved. Nonzero integer Fourier indices under $A_a^\top$ never repeat: a repetition would give a root of unity on their rational cyclic subspace. Distinct integer vectors eventually leave each bounded set. Character correlations therefore vanish eventually for each fixed pair of nonconstant characters. Approximation by trigonometric polynomials and unitarity prove mixing on L^2 . On the real central plane, diagonalizability gives $\sup_{n \in \mathbb{Z}} \|A_a^n v\| \leq C_a \|v\|$. Arbitrarily small nonzero v project to points remaining arbitrarily close to zero for all time. Hence the map is not expansive. $\square$

2 Every fixed group, return and zeta coefficient

Theorem 1. For every $n \geq 1$, with $B_n = A_a^n - I$, the complete fixed group is

$$\text{Fix}(A_a^n) \simeq \mathbb{Z}^4 / B_n \mathbb{Z}^4 \simeq \bigoplus_{j=1}^4 \mathbb{Z} / d_j \mathbb{Z},$$

where $d_1 \mid \dots \mid d_4$ are the positive Smith factors of B_n . Its cardinality and the number O_n of oriented primitive cycles are

$$F_n = (\lambda^n + \lambda^{-n} - 2)(2 - 2 \cos n\theta), \quad (1)$$

$$nO_n = \sum_{d|n} \mu(n/d) F_d. \quad (2)$$

For $|z| < \lambda^{-1}$ the ordinary zeta converges absolutely, and its complete rational continuation is

$$Z_a(z) = \exp \sum_{n \geq 1} \frac{F_n z^n}{n} = \prod_{n \geq 1} (1 - z^n)^{-O_n} = \frac{(1 - z)^4 Q_a(z)}{P_a(z)^2}, \quad (3)$$

where $Q_a(z) = 1 + 3z + (a^2 + 4)z^2 + 3z^3 + z^4$.

Proof. No eigenvalue is a root of unity, so B_n is nonsingular. Its kernel on the torus is $B_n^{-1} \mathbb{Z}^4 / \mathbb{Z}^4$; multiplication by B_n identifies it with the stated cokernel. Smith reduction gives both its structure and its order. The real expanding and contracting factors of $\det(A_a^n - I)$ have opposite signs; the conjugate pair has positive product. Thus $\det(A_a^n - I) = -F_n$, which gives (1). Partitioning fixed points by least period gives $F_n = \sum_{d|n} dO_d$ and Möbius inversion gives (2). This proves nonnegative integrality of O_n without a numerical test.

Every return has derivative A_a^n , hence multipliers $\lambda^n, \lambda^{-n}, e^{in\theta}, e^{-in\theta}$. A primitive cycle of length d contributes its d marked points at every repetition. Reversal preserves length and reciprocates multipliers; we do not divide self-reversing or paired cycles by two. The unweighted cardinality convention has no extra phase; its signed Lefschetz counterpart is $-F_n$, not F_n .

Since $F_n \leq 4\lambda^n$, the initial series is absolutely convergent. Use $\det(I - A_a^n) = \sum_{j=0}^4 (-1)^j \text{tr}((\wedge^j A_a)^n)$ and $\det(I - z \wedge^2 A_a) = (1 - z)^2 Q_a(z)$. Exterior degrees one and three both give $P_a(z)$, and zero and four both give $1 - z$. The negative cardinality/index sign puts even exterior degrees in the numerator, proving (3). $\square$

The Koopman operator itself is unitary on infinite-dimensional L^2 , so it is noncompact and is not the trace-class operator of an ordinary Fredholm determinant. Formula (3) makes no such identification.

3 Primitive and cumulative fluctuation laws

Theorem 2. For fixed admissible a ,

$$\frac{nO_n}{\lambda^n} = 2 - 2 \cos(n\theta) + O_a(n\lambda^{-n/2}). \quad (4)$$

The left side has cluster set $[0, 4]$, limiting density $1/(\pi\sqrt{x(4-x)})$ on $(0, 4)$, mean 2 and variance 2. For $\Pi(N) = \sum_{n \leq N} O_n$, $r = \lambda^{-1}$, put

$$C = \frac{2}{1-r}, \quad B = \frac{2}{|1-re^{-i\theta}|}.$$

Then $C > B > 0$ and

$$\frac{N\Pi(N)}{\lambda^N} = C - 2\Re \frac{e^{iN\theta}}{1-re^{-i\theta}} + O_a(N^{-1}). \quad (5)$$

Its cluster set is $[C-B, C+B]$ and its limiting density is $1/(\pi\sqrt{B^2 - (x-C)^2})$. No constant K gives $\Pi(N) \sim K\lambda^N/N$.

Proof. Every proper divisor of n is at most $n/2$. In (2), use $F_d \leq 4\lambda^d$ and at most n divisors. In addition, $F_n/\lambda^n - (2-2\cos n\theta) = O_a(\lambda^{-n})$. This proves the absolute, not relative, error in (4). For each nonzero integer k , the Cesàro average of $e^{ikn\theta}$ tends to zero by its geometric-sum formula. Trigonometric approximation proves equidistribution on the circle. Pushing uniform angle through $2-2\cos t$ gives the displayed density, cluster set and moments. No assertion of small relative error is made near its zero endpoint.

To sum (4), write $n = N - j$. For $0 \leq j \leq N/2$, $N/(N-j) = 1 + O(j/N)$; the sum weighted by r^j has error $O_a(1/N)$. Terms $n < N/2$ and the omitted geometric tails are exponentially small after normalization, bounded by $O_a(N^2\lambda^{-N/2})$. Summing r^j and $(re^{-i\theta})^j$ gives (5). Since $\theta \notin 2\pi\mathbb{Z}$, $|1-re^{-i\theta}| > 1-r$. The same equidistribution argument gives the translated arcsine law. Its nonzero width rules out a constant prefactor. The proof never uses a lower bound on $|e^{in\theta} - 1|$. $\square$

4 Verification and the arithmetic boundary

The independent audit uses $a = 1, 3, 4, 5, 8$ and $1 \leq n \leq 24$: 120 complete finite fixed groups, 1,920 return-matrix cells, and twelve cyclotomic controls. A permutation/minor producer is checked by Newton power sums and independent Smith reduction. Symbolic symplectic/reversal and exterior identities are supplemented by 70-digit root calculations. These are finite regression checks, not proofs of equidistribution or homoclinic triviality. Full domains and infinite quantifiers are proved above.

The native algebraic unit, integer fixed groups and original cycle clock justify only weak arithmetic/orbit evidence. No intrinsic rational-prime carrier, logarithmic prime roof or target divisor relation is present. The conservative tuple is weak A0, weak A1, failed A2/A3 and a formal A4 hint; overall the target candidate is rejected. This verdict does not invalidate the source theorems. All nine target/Route-B flags are false: NO_BAD_EULER_OR_ROOT_NUMBER.

Revision record. Round one adds complete primitive and cumulative fluctuation laws.

References

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